

Assignment 2

(a) You have a light source that is too bright, so you want to illuminate it at **50%** of its total illumination capacity. **Design** a square wave generator that accomplishes this by using only **12V and -8V voltage sources, 3 resistors and 1 capacitor**. Additionally, you must be careful such that any light flickering is **unnoticeable to the human eye** (Hint: Human eyes can notice flickering up to **500 Hz**). You must satisfy all the mentioned criteria, and assume any component values accordingly. **[10 marks]**

(a)	Draw the complete circuit of the square wave generator
(b)	Determine all the component values (resistors, capacitors, saturation voltages)
(c)	Plot the time vs output voltage curve (Time vs V_{in}/V_{out} plot) . Clearly label the plot with calculated values

(b) **Design** a square wave generator with **30% Duty Cycle**. **[10 marks]**